

JAWAD ZAMAN

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EDUCATION

St. Joseph’s University New York Bachelor of Science (Honors), Triple Major Computer Science • Mathematics • Business Administration GPA: 4.00 Dean’s List (2024-2026) Management Certificate (2026) • Leadership and Supervision Certificate (2026)	Brooklyn, New York Expected May 2027
Lakeland University Japan Study Abroad Exchange Program GPA: 4.00 Dean’s Honors List	Tokyo, Japan Fall 2025

PROFESSIONAL EXPERIENCE

New York Academy of Sciences Scientist-in-Residence (SIR) •Design and deliver project-based Arduino robotics curriculum for 120+ NYC high school students, with materials adopted by the SIR program •Mentor high school seniors through technical projects, fostering engineering skills and career readiness in robotics and AI systems •Guide students to successfully present their original robotics projects at the annual New York Academy of Sciences showcase	New York, New York August 2025–Present
St. Joseph’s University New York Academic Center Tutor •Tutor up to 20 hours weekly across advanced Computer Science, Mathematics, Economics, and Business coursework •Deconstruct complex topics like data structures, calculus, and financial modeling into structured workflows •Deliver individualized instruction to enhance student comprehension, exam readiness, and course retention rates	Brooklyn, New York August 2026–Present
Ecological Society of America Treasurer •Manage international finances for membership dues, conference registrations, and research initiatives supporting 5,000+ members globally •As an ESA Certified Ecologist, co-organize the Annual Meeting, coordinating logistics, programming, and engagement for researchers •Collaborate with executive leadership on budget planning and resource allocation for scientific programs and international collaborations	Washington, D.C. October 2025–Present
Adamjee Cantonment College Robotics Club Director of Olympiads •Orchestrated national-level STEM Olympiads in robotics, mathematics, and sciences for 500+ participants, managing logistics and judging panels •Increased medal count by 35% through curriculum-aligned training modules, mentoring sessions, and strategic university partnerships •Collaborated with external partners to align events with national education standards, enhancing student readiness and competitive performance	Dhaka, Bangladesh January 2023–December 2024

FIRST AUTHOR PUBLICATIONS

“Benchmarking Neural Machine Translation Using Open-Source Transformer Models and a Comparative Study with a Focus on Medical and Legal Domains” NRHC Conference 2025 NRHC <i>Illuminate</i> Journal Volume 7, Fall 2025 [link] •Authored a first author study evaluating open-source transformer models on medical and legal texts to reduce translation errors •Won the NRHC Paper of the Year award from thousands of submissions and received the Student of the Year honorable mention •Secured multiple undergraduate research support grants to present these findings at the annual conference in Harrisburg, Pennsylvania
“Healithic: A Patient-Controlled, Blockchain-Secured Electronic Medical Record System for Decentralized Healthcare in Bangladesh” NRHC <i>Illuminate</i> Journal Volume 8, Fall 2026 [link] •Inspired by designing an EMR for a local hospital, this research addresses patient autonomy gaps in Bangladesh’s healthcare system •Developed a blockchain secured, patient controlled EMR system using smart contracts, wallet authentication, and encrypted off chain storage •Accepted for publication in NRHC <i>Illuminate</i>; functional prototype publicly deployed via GitHub Pages for community testing
“Emotions in the Eye of the Machine: Accuracy, Bias, and Ethical Implications of AI-Based Facial Analysis” NRHC Conference 2026 / <i>IEEE Transactions on Affective Computing</i> (Under peer review) [link] •Led a study evaluating three transformer models on six thousand facial images to identify demographic accuracy disparities •Presented findings with a live demonstration at the NRHC Conference, earning the Student of the Year honorable mention •Funded by the Scholarship of Equity and Access, undergraduate research grant, and honors research grant; under IEEE peer review
“Spatiotemporal Decoding of Dynamic Risk Decisions using Machine Learning and Functional MRI” <i>IMPULSE: The Premier Undergraduate Neuroscience Journal</i> (Accepted) [link] •Trained a PyTorch neural network on functional MRI data from sixteen participants to decode sequential risk taking decisions •Achieved sixty four percent decoding accuracy, identifying the Frontal Opercular Cortex and Subcallosal Cortex as key predictive regions •Accepted for publication in IMPULSE, the premier undergraduate neuroscience journal, showcasing multivariate pattern analysis of brain activity
“Honors as Culminating Experience for Computer Science Majors in the AI Era” <i>Journal of the National Collegiate Honors Council</i> (Accepted) •Conducted an IRB approved survey of one hundred six honors students and faculty to assess AI era career readiness •Revealed a disconnect between faculty emphasis on ethical inquiry and student preference for practical, project based AI preparation •Accepted for publication in the Journal of the National Collegiate Honors Council; IRB approved methodology ensured ethical compliance

PUBLISHED CONFERENCE ABSTRACTS

“NASA-powered hyperspectral machine learning with NEON and Landsat imagery to predict vegetation traits at multiple spatial resolutions in Harvard Forest and Black Rock Forest” Ecological Society of America (ESA) 2026 Annual Meeting Abstracts [link]
“Enhancing NASA GEDI forest biomass estimates through spatially-stratified machine learning” Ecological Society of America (ESA) 2026 Annual Meeting Abstracts [link]
“Comparing Machine Learning Models for Environmental and Ecological Monitoring using Remote Sensing Data in the Midwest and Northeast” Ecological Society of America (ESA) 2025 Annual Meeting Abstracts [link]

- “From Joy to Hate: A Comparative Analysis in Evaluating Emotion-Specific Accuracy in Transformer-Based AI” | CURLI 2026 Annual Undergraduate STEM Conference Proceedings [\[link\]](#)
- “Spatially-stratified machine learning frameworks for refined NASA GEDI forest biomass mapping” | CURLI 2026 Annual Undergraduate STEM Conference Proceedings [\[link\]](#)
- “Spectral Signatures and Smart Algorithms for Tracking Vegetation Health Across Forest Sites in Massachusetts and New York” | The 13th Annual Student Research Symposium Proceedings, St. Joseph’s University, New York [\[link\]](#)
- “Benchmarking Neural Machine Translation Using Open-Source Transformer Models: A Comparative Study with a Focus on Medical and Legal Domains” | The 12th Annual Student Research Symposium Proceedings, St. Joseph’s University, New York [\[link\]](#)
- “Performance Evaluation of Remote Sensing-Driven Machine Learning Models in Midwestern and Black Rock Forest Ecosystems” | The 12th Annual Student Research Symposium Proceedings, St. Joseph’s University, New York [\[link\]](#)

RESEARCH PRESENTATIONS

- “Evaluating a Post hoc Concept Bottleneck Model for Multi Modal Feature Weighting in Skin Cancer” | NRHC Conference 2027 (Accepted) [\[link\]](#)
Developed an interpretable AI system utilizing concept bottleneck models and large language models to generate clinical diagnostic reports
- “Comparing Machine Learning Models for Environmental and Ecological Monitoring using Remote Sensing Data” | ESA Annual Meeting 2025
Trained decision tree and random forest models on satellite imagery to accurately predict vegetation health and tree diameter metrics
- “Predictive Modeling for Financial Solvency” | Islamic University of Technology Applied Machine Learning Research Symposium 2024
Won third prize by classifying mobile device price ranges and financial solvency using extensive data preprocessing and machine learning models
- “Mechanistic Evolution of Digital Wireheading and Artificial Reward Addiction in Reinforcement Learning Agents” | SJNY Honors Symposium
Simulated reward hacking behaviors in reinforcement learning agents to evaluate artificial addiction metrics and digital wireheading mechanisms
- “Hyperspectral Machine Learning with NEON and Landsat Imagery to Predict Vegetation Traits” | ESA Annual Meeting 2026
Applied principal component analysis and gradient boosting algorithms to predict vegetation indices across Harvard Forest and Black Rock Forest
- “Enhancing NASA GEDI Forest Biomass Estimates through Spatially Stratified Machine Learning” | CURLI Conference 2026
Improved satellite biomass predictions using random forests and geographic cross validation across three thousand Pacific Northwest sites
- “Spectral Signatures and Smart Algorithms for Tracking Vegetation Health Across Forest Sites” | SJNY Research Symposium 2025
Compared multispectral and hyperspectral satellite data using machine learning to evaluate canopy density and greenness differences
- “Machine Learning Classifiers for Quantifying Particulate Microplastic Contamination in Urban Aquatic Systems” | NYC Microplastic Symposium
Applied machine learning over multiple years to track environmental shifts and quantify particulate contamination across urban aquatic ecosystems
- “Intercultural Communications and German Socio-Cultural Analysis” | Lakeland University World Expo 2025
Presented an interactive booth exploring the unique intersections of German business practices and socio-cultural influences within modern Japan

LEADERSHIP EXPERIENCE

President Student Government Association, St. Joseph’s University New York •Lead all student organizations and approve budgets, representing SJNY at the 2026 ASGA National Conference in Washington D.C. •Revitalize the Campus Activity Board by recruiting and onboarding new officers to drive programming and student engagement •Received the Outstanding Student Organization Leader Award at the 2026 Student Leadership Gala for sustained organizational impact	May 2026–Present
Chairperson SGA Food Pantry Committee, St. Joseph’s University New York •Lead initiatives to combat campus food insecurity by managing inventory logistics and coordinating weekly distribution schedules •Fundraise for the Joseph’s Hands grant and organize volunteering trips to food pantries in Chicago and St. Louis •Maintain inventory management systems ensuring consistent supply availability for students experiencing food insecurity on campus	January 2025–Present
President & Founder Honors Student Committee, St. Joseph’s University New York •Founded the university’s first student-led Honors Committee and authored its governing constitution and bylaws •Mentor honors students through early-stage research ideation, methodology design, and undergraduate thesis preparation •Managed digital outreach, official club communications, and social platforms to drive student engagement	September 2026–Present
President Go Green Club, St. Joseph’s University New York •Raised \$3,000 at SJNY Giving Day to convert an empty space into the university’s first courtyard •Led sustainability initiatives and environmental awareness campaigns, earning Student Organization of the Year in both 2025 and 2026 •Received the Emerging Leader Award at the 2025 Student Leadership Gala for campus sustainability leadership	May 2024–May 2026
President Book Club, St. Joseph’s University New York •Transformed a dormant club into one of the most active organizations by managing literary events and community reading programs •Founded the Book Club Mini Library, creating a permanent student resource for accessible reading materials on campus •Doubled the annual budget allocation from \$500 to \$1000 through demonstrated programming growth	May 2024–May 2026
Treasurer Computer Science Club, St. Joseph’s University New York •Designed and launched the first official Computer Club website, establishing a centralized digital hub for events and member resources •Oversaw club finances and co-organized university wide tech talks and coding workshops, inviting speakers from Microsoft, NASA, and more •Fundraised approximately \$1,000 to support the club’s annual tech talk, coding workshops, and professional development events	May 2024–May 2026

Panel Moderator

March 2025

Northeast Regional Honors Council (NRHC)

Selected to lead academic discourse and moderate research panels for honors students across the Northeast region

Facilitated structured question and answer sessions connecting presenters with faculty reviewers and attending honors delegates

Coordinated session timing and thematic transitions to ensure equitable participation across multiple concurrent research presentations

CLINICAL EXPERIENCE

ZWH Medical Care PC

Woodside, New York

Clinical Volunteer

May 2024–August 2026

Accumulated 1,000+ clinical hours of direct patient contact every summer, collecting chief complaints, medical histories, and vital signs

Observed physician patient encounters and diagnostic decision making, gaining exposure to disease management and symptom presentation

Partnered with physicians and clinical staff to manage high volume outpatient workflows, EHR documentation, and laboratory tracking systems

The Brooklyn Hospital Center

Brooklyn, New York

Student Volunteer

December 2024

Observed busy holiday patient intake and triage workflows, gaining exposure to emergency and acute care operational dynamics

Assisted nursing and administrative staff with ancillary logistical support to maintain efficient care delivery during peak holiday census

Engaged with diverse patient populations in a dynamic urban hospital setting, reinforcing foundational bedside communication skills

AWARDS & HONORS

NRHC “Paper of the Year” (2025) | Top research recognition across Northeast U.S. honors institutions for medical AI translation studies

NRHC “Student of the Year” Honorable Mention (2025, 2026) | Recognized twice for outstanding academic, research, and ethical leadership

NRHC “Scholarship of Equity & Access” (2025, 2026) | Two-time awardee selected from hundreds of applicants for ethics-focused AI research

SJNY Undergraduate Research Grant (2024-2026) | Three-time recipient funding travel and presentations for major national conferences

NRHC Honors Conference Travel Grant (2025, 2026) | Fully funded registration and lodging to present artificial intelligence research papers

NASA Space Apps “Global Nominee” (2024, 2025) | Ranked in the top 0.5% of over 30,000 global participants for space technology hackathons

Xylem Global Innovation Challenge First Place (2025) | Led RESQMAP flood-rescue AI team among eight hundred global university teams

Columbia Ivy League DevFest Hackathon Winner (2025) | Built Forager on the Go, an AI species-identification app for urban nature tracking

ACM Hackathon Champion (2025) | Led the rapid development of an immersive 3D video game using Unity and C# in a highly timed competition

Stony Brook HopperHacks Best Project (2025) | Won the Treasure Trove of Talent strand for NeuroNudge, an innovative AI communication tool

NYAS Certificate of Excellence (March 2026) | Awarded to the top-performing Scientist-in-Residence for exceptional STEM mentorship skills

SJNY “Established Leader Award” (2026) | Highest student government honor for revitalizing campus activity boards and managing clubs

“Outstanding Student Org Leader” Award (2025) | Honored for exceptional leadership across the Go Green, Book, and Computer Science clubs

“Joseph’s Hands” Grant Recipient (2025) | Secured \$1,000 to combat campus food insecurity and fund student pantry logistics

IUT Applied ML Research Symposium Third Prize (2024) | Recognized for predictive modeling of financial solvency using hardware specs

Father Timm Memorial Programming Contest 3.0 (2024) | Placed 13th overall in a rigorous and competitive national algorithmic coding contest

TECHNICAL SKILLS

AI & ML: PyTorch, TensorFlow, Scikit-Learn, OpenCV, YOLO, NLTK, spaCy, Keras, Computer Vision, NLP, Bias Mitigation, Random Forest

Languages: Python, JavaScript, TypeScript, C/C++, C#, Java, SQL, R, Solidity, Bash, PowerShell, HTML5, CSS3, Assembly, MATLAB, Go, JSON, XML

Web & DApps: React, Next.js, Node, Flask, Express, Tailwind, Web3.js, Ethers.js, REST API, Groq, GitHub Pages, Vercel, DApps, U^2-Net, Rembg

Geospatial & GIS: Rasterio, NEON HSI, Landsat 8, SRTM DEM, Google Earth Engine, GeoTIFF, PCA, LiDAR, SHAP, Spatial CV, QGIS, NASA APIs, ORNL

Neuro & Biostats: Nilearn, BIDS, NifTI, fMRI, Clopper-Pearson, Fisher’s Exact Test, SciPy, StatsModels, Brier Score, MVPA, Harvard-Oxford Atlas

Data Science: Pandas, NumPy, Matplotlib, Seaborn, SciPy, Rasterio, Google Earth Engine, NEON Data, NASA APIs, Jupyter, Colab, Plotly, Tableau

Hardware & Games: Unity, C#, Arduino, WebGL, Three.js, Sensors, Microcontrollers, Robotics, IoT, Breadboarding, ESP32, Raspberry Pi, Circuits

MEDIA & PRESS COVERAGE

SJNY News (Nov 2025): Profiled for winning the NRHC Best Paper of the Year award. [\[link\]](#)

SJNY News (Mar 2025): Featured for double first-place hackathon wins at Columbia and Stony Brook. [\[link\]](#)

SJNY News (Sep 2026): Highlighted for delivering the Student Pledge at the university Investiture Ceremony. [\[link\]](#)

SJNY News (Mar 2026): Quoted as Go Green Club President on cross-campus sustainability initiatives. [\[link\]](#)

SJNY News (Sep 2026): Interviewed as an orientation leader on community building and student mentorship. [\[link\]](#)

SJNY News (May 2026): Featured for undergraduate poster presentation and research contributions at the symposium. [\[link\]](#)

SJNY News (Aug 2025): Quoted on service, purpose, and community leadership in Kansas City. [\[link\]](#)

SJNY News (May 2025): Highlighted for podium oral presentation evaluating neural machine translation models. [\[link\]](#)

SJNY News (May 2025): Recognized for academic distinction and induction into Sigma Iota Chi. [\[link\]](#)

SJNY News (Mar 2025): Profiled during Eastern District Court visit exploring civic pathways and justice. [\[link\]](#)

SJNY News (Apr 2025): Spotlighted for community outreach and food pantry service in Chicago. [\[link\]](#)

ACES Magazine (Nov 2025): Authored feature on global scholarship and study-abroad experiences in Tokyo. [\[link\]](#)

SurajC Blog (Mar 2025): Profiled in a personal essay on authentic leadership and campus connections. [\[link\]](#)

ACADEMIC REFERENCES

Dr. Elizabeth Zollinger, Ph.D. | Chair & Associate Professor of Math & CS, SJNY

Email: ezollinger@sjny.edu | Phone: (718) 940-5758

Dr. Anna Rita Pennisi, Ph.D. | Assistant Professor of Psychology & Honors Program Director, SJNY

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Dr. Michael Burke, Ph.D. | Professor of Philosophy & Associate Dean of Student Success, SJNY

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Dr. David Seppala-Holtzman, Ph.D. | Professor Emeritus of Mathematics & Computer Science, SJNY

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